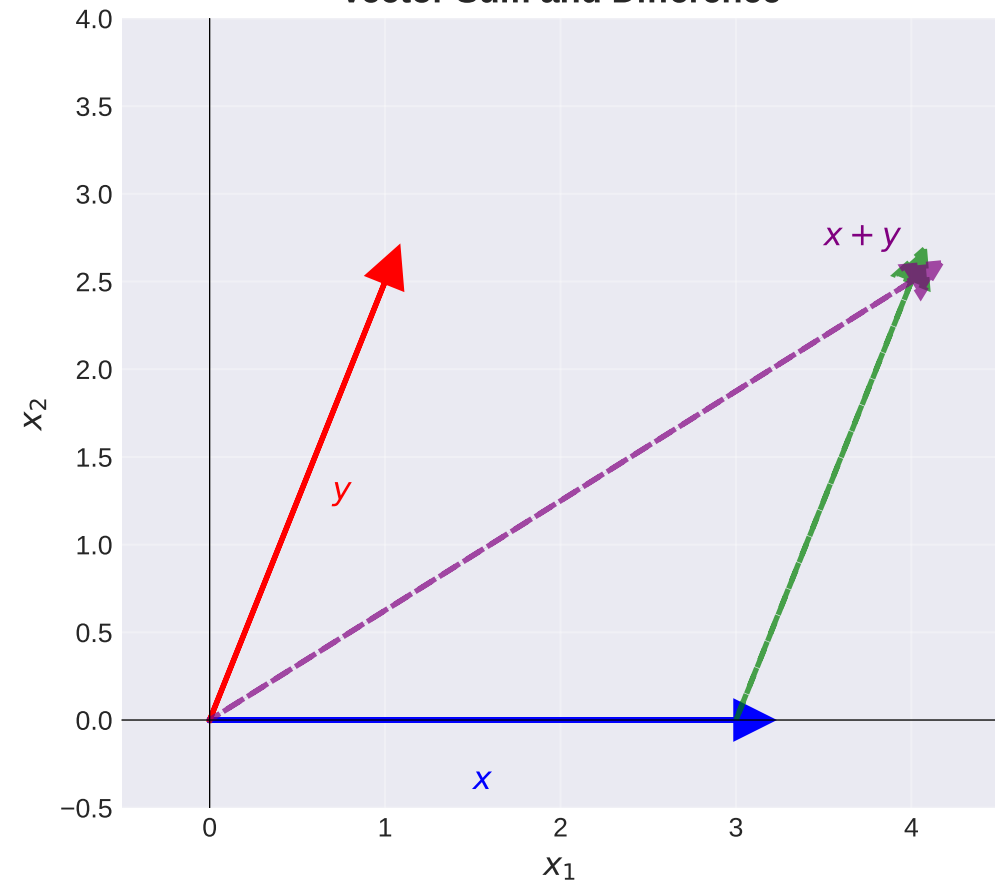


Vector Sum and Difference



Parallelogram Law

$$2(\|x\|^2 + \|y\|^2) = \|x + y\|^2 + \|x - y\|^2$$

*This law characterizes inner product spaces.
A normed space is an inner product space
if and only if the parallelogram law holds.*